

DAC_phase 4: DEVELOPMENT PART-2

```
import pandas as pd
```

```
traffic_data = pd.read_csv(r"C:\\Users\\tarun\\OneDrive\\Desktop\\ daily-website-visitors.csv")
```

```
print(traffic_data.head())
```

OUTPUT:

Row	Day	Day.Of.Week	Date	Page.Loads	Unique.Visits
0	1	Sunday	1 9-14-2014	2,146	1,582
1	2	Monday	2 9-15-2014	3,621	2,528
2	3	Tuesday	3 9-16-2014	3,698	2,630
3	4	Wednesday	4 9-17-2014	3,667	2,614
4	5	Thursday	5 9-18-2014	3,316	2,366

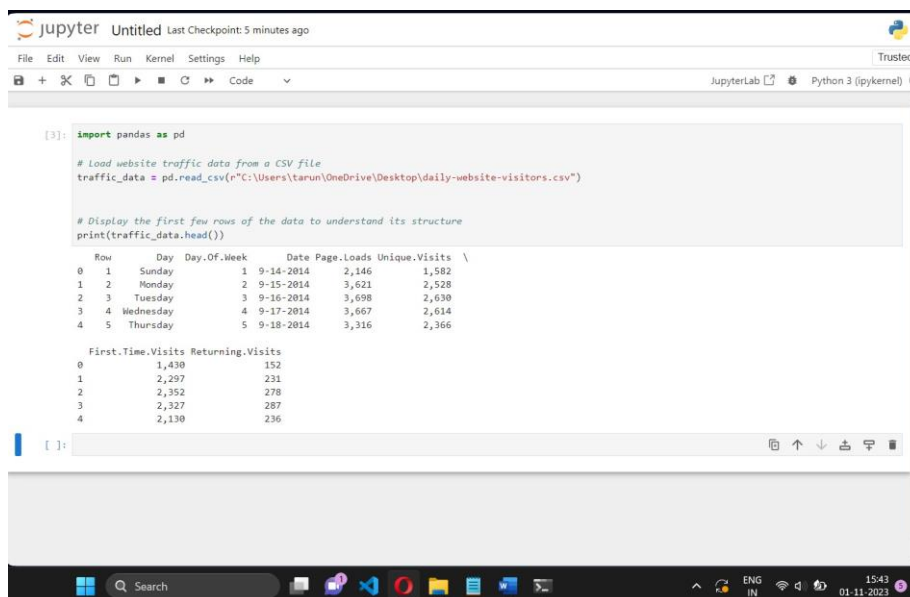
	First.Time.Visits	Returning.Visits
0	1,430	152
1	2,297	231
2	2,352	278
3	2,327	287
4	2,130	236

```
import pandas as pd
```

```
traffic_data = pd.read_csv(r"C:\\Users\\tarun\\OneDrive\\Desktop\\ daily-website-visitors.csv")
```

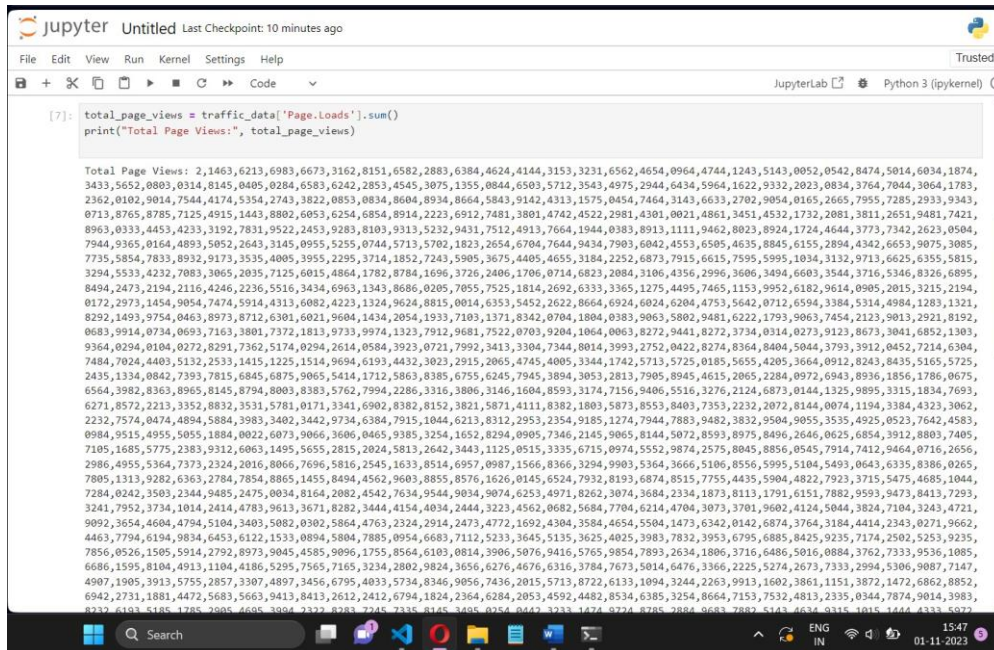
```
print(traffic_data.head())
```

OUTPUT IMAGE:



total_page_views=traffic_data['Page.Loads'].sum()

print("Total page views:",total_page_views)

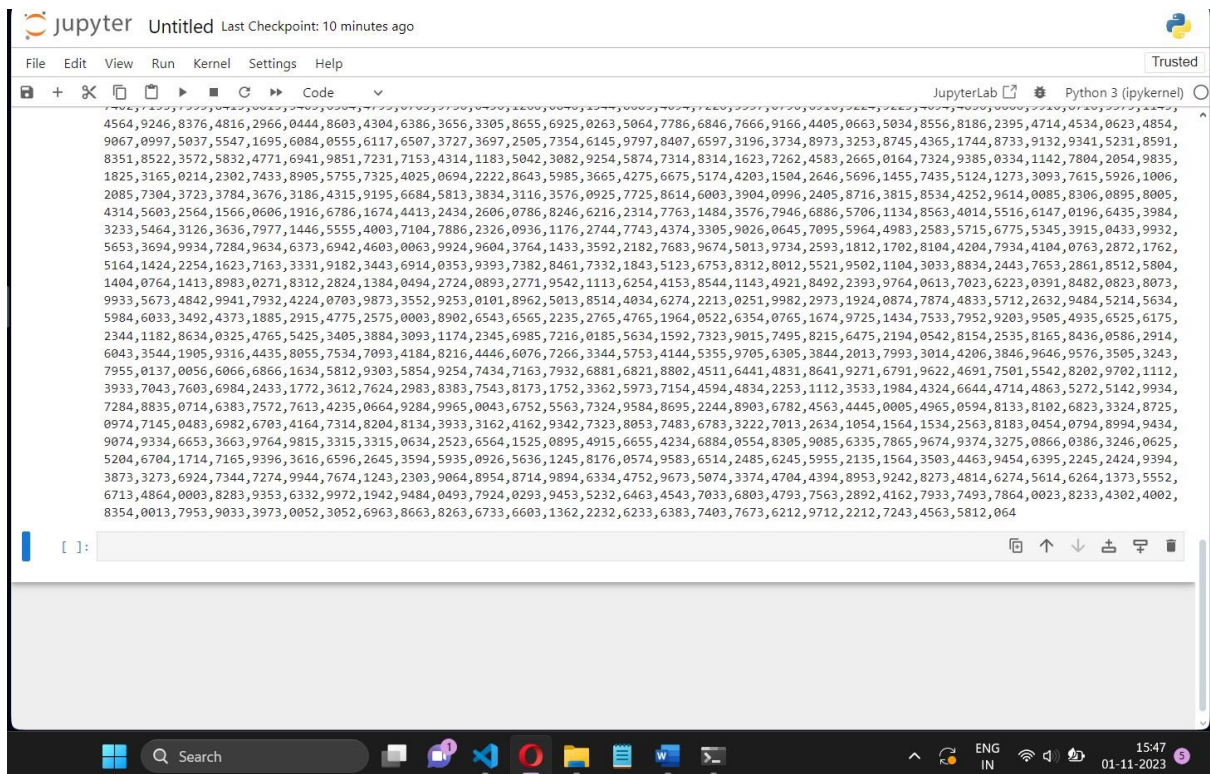


The image shows a JupyterLab interface with a code cell. The code cell contains the following Python code:

```
[7]: total_page_views = traffic_data['Page.Loads'].sum()
print("Total Page Views:", total_page_views)
```

The output of the code cell is a large integer representing the total page views:

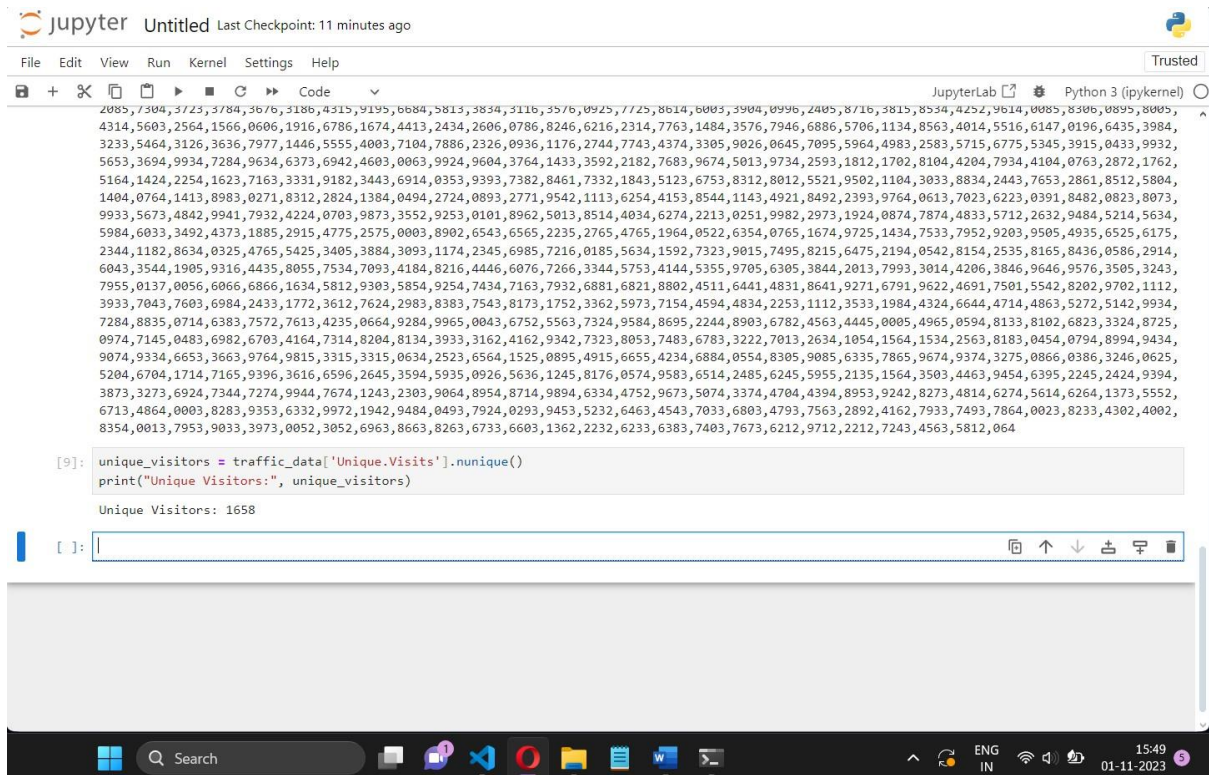
```
Total Page Views: 2,1463,6213,6983,6673,3162,8151,6582,2883,6384,4624,4144,3153,3231,6562,4654,0964,4744,1243,5143,0052,0542,8474,5014,6034,1874,
3433,5652,0803,0314,8145,0405,0284,6583,6242,2853,4545,3075,1355,0844,6503,5712,3543,4975,2944,6434,5964,1622,9332,2023,0834,3764,7044,3064,1783,
2362,0102,9014,7544,4174,5354,2743,3822,0853,0834,8604,8934,8664,5843,9142,4313,1575,0454,7464,3143,6633,2702,9054,0165,2665,7955,7285,2933,9343,
0713,8765,8785,7125,4915,1443,8802,6053,6254,6854,8914,2223,6912,7481,3801,4742,4522,2981,4301,0021,4861,3451,4532,1732,2081,3811,2651,9481,7421,
8963,0333,4453,4233,3192,7831,9522,2453,9283,8103,9313,5232,9431,7512,4913,7664,1944,0383,8913,1111,9462,8023,8924,1724,4644,3773,7342,2623,0504,
7944,9365,0164,4893,5052,2643,3145,0055,5255,0744,5713,5702,1823,2654,6704,7644,9434,7903,6042,4553,6505,4635,8845,6155,2804,4342,6653,9075,3005,
7735,5854,7833,8932,9173,3535,4805,3955,2295,3714,1852,7243,5905,3675,4405,4655,3184,2252,6873,7915,6615,7595,5995,1034,3132,9713,6625,6355,5815,
3294,5533,4232,7083,3065,2935,7125,6015,4864,1782,8784,1696,3726,2406,1706,0714,6823,2084,3106,4356,2996,3606,3494,6603,3544,3716,5346,8126,6895,
8494,2473,2194,2116,4246,2236,5516,3434,6963,1343,8686,0205,7055,7525,1814,2692,6333,3365,1275,4495,7465,1153,9952,6182,9614,0905,2015,3215,2194,
0172,2973,1454,9054,7474,5914,4313,6082,4223,1324,9624,8815,0014,6353,5452,2622,8664,6924,6024,6204,4753,5642,0712,6594,3384,5314,4984,1283,1321,
8292,1493,9754,0463,8973,8712,6301,6021,9604,1434,2054,1933,7103,1371,8342,0704,1804,0383,9063,5802,9481,6222,1793,9063,7454,2123,9013,2921,8192,
0683,9914,0734,0693,7163,9733,9974,1323,7912,9681,7522,0703,9204,1064,0063,8272,9441,8272,3734,0314,0273,9123,8673,3041,6852,1304,
9364,0294,0104,0272,8291,7362,5174,0294,2614,0584,3923,0721,7992,3413,3304,7344,8014,3993,2752,0422,8274,8364,8404,5044,3793,3912,0452,7214,6304,
7484,7024,4403,5132,2533,1415,1125,1514,9694,6193,4432,3023,2915,2065,4745,4005,3344,1742,5713,5725,0185,5655,4205,3664,0912,8243,8435,5165,5725,
2435,1334,0842,7393,7815,6845,6875,9065,5414,1712,5863,8385,6755,6245,7945,3894,3053,2813,7905,8945,4615,2065,2284,0972,6943,8936,1856,1786,0675,
6564,3982,8363,8965,8145,8794,8003,8383,5762,7994,2286,3316,3806,3146,1604,8593,3174,7156,9406,5516,3276,2124,6873,0144,1325,9895,3315,1834,7693,
6271,8572,2213,3352,8832,3531,5781,0171,3341,6902,8382,8152,3821,5871,4111,8382,1803,5873,8553,8403,7353,2232,2072,8144,0074,1194,3384,4323,3062,
2232,7574,0474,4894,5884,3983,3402,3442,9734,6384,7915,1044,6213,8312,2953,2354,5185,1274,7944,7883,9482,3832,9504,9055,3335,4025,0523,7642,4583,
0894,9515,4955,5055,1884,0022,6073,9606,3606,0445,9385,3254,1652,8294,0905,7346,2145,9065,8144,5072,8593,8975,8496,2646,0625,6854,3912,8803,7405,
7105,1685,5775,2383,9312,6063,1495,5655,2815,2024,5813,2642,3443,1125,0515,3335,6715,0974,5552,9874,2575,8045,8856,8545,7914,7412,5464,0716,2656,
2986,4955,5364,7373,2324,2016,8066,7696,5816,2545,1633,8154,6957,0987,1566,8366,3294,9903,5364,3666,5106,8556,5995,5104,5493,0643,6335,8386,0265,
7805,1313,9282,6363,2784,7854,8865,1455,8494,4562,9603,8855,8576,1626,0145,6524,7932,8193,6874,8515,7755,4435,5904,4822,7923,3715,5475,4685,1044,
7284,0242,3503,2344,9485,2475,0034,8164,2082,4542,7634,9544,9034,9074,6253,4971,8262,3074,3684,2334,1873,8113,1791,6151,7882,9593,9473,8413,7693,
3241,7952,3734,1014,2414,4783,9613,3671,8282,3444,4154,4034,2444,3223,4562,0682,5684,7704,6214,4704,3073,3701,9602,4124,5044,3824,7104,3243,4721,
9092,3654,4604,4794,5104,3403,5082,0302,5864,4763,2324,2914,2473,4772,1692,4304,3584,4654,5504,1473,6342,0142,6874,3764,3184,4414,2343,0271,9662,
4463,7794,6194,9834,6452,6122,1533,0894,5884,7885,0954,6683,7112,5233,3645,5135,3625,4025,3983,7832,3953,6795,6885,8425,9235,7174,2502,5525,9235,
7856,0626,1505,5914,2792,8973,9045,4585,9096,1755,8564,6103,0814,3906,5076,9416,5765,9854,7893,2634,1806,3716,6486,5016,0884,3762,7333,9536,1005,
6686,1595,8104,4913,1104,4186,5295,7565,7105,3234,2802,9824,3656,6276,6476,6316,3784,7673,5014,6476,3366,2225,5274,2673,7333,2994,5306,9087,7147,
4907,1965,3913,5755,2857,3307,4897,3456,6795,4833,5734,8346,9856,7436,2015,5713,8722,6133,1094,3244,2263,9913,1602,3861,1151,3872,1472,6862,0852,
6942,2731,1881,4472,5683,5663,9413,8413,2612,2412,6794,1824,2364,6284,2053,4592,4482,8534,6385,3254,8664,7153,7532,4813,2335,0344,7874,9014,3983,
8332,6103,5184,1478,3906,3605,3904,2332,8283,7245,7335,8145,3495,8254,0447,3213,1474,8774,8786,3884,8663,7283,5143,4634,8315,1015,1444,4333,5072,
```



The image shows a JupyterLab interface with a code cell. The code cell contains a large list of integers, which appears to be a continuation of the output from the previous code cell. The integers are displayed in a single line, wrapped across the width of the code cell. The list starts with 4564 and ends with 8354,0013,7953,9033,3973,0052,3052,6963,8663,8263,6733,6603,1362,2232,6233,6383,7403,7673,6212,9712,2212,7243,4563,5812,064.

```
unique_visitors=traffic_data['Unique.Visits'].nunique()
```

```
print("Unique Visitors:",unique_visitors)
```



The screenshot shows a JupyterLab window titled 'Untitled' with a 'Last Checkpoint: 11 minutes ago' status. The interface includes a top bar with 'File', 'Edit', 'View', 'Run', 'Kernel', 'Settings', and 'Help' menus. Below the top bar is a toolbar with icons for file operations and a 'Code' dropdown menu. The main area contains a code cell with the following text:

```
[9]: unique_visitors = traffic_data['Unique.Visits'].nunique()
print("Unique Visitors:", unique_visitors)
```

The output of the code cell is displayed below the code:

```
Unique Visitors: 1658
```

The bottom of the screenshot shows a Windows taskbar with a search bar, several application icons, and a system clock indicating 15:49 on 01-11-2023.

Analyze traffic sources:

```
traffic_sources = traffic_data['Page.Loads'].value_counts()
```

```
print("Traffic Sources:\n", traffic_sources)
```

```
# Visualize the data (you may need to install additional libraries for this)
```

```
import matplotlib.pyplot as plt
```

```
# Example: Plot page views over time
```

```
traffic_data['Date'] = pd.to_datetime(traffic_data['Date'])
```

```
traffic_data.set_index('Date', inplace=True)
```

```
traffic_data['Page.Loads'].plot(title='Page Views Over Time')
```

```
plt.show()
```

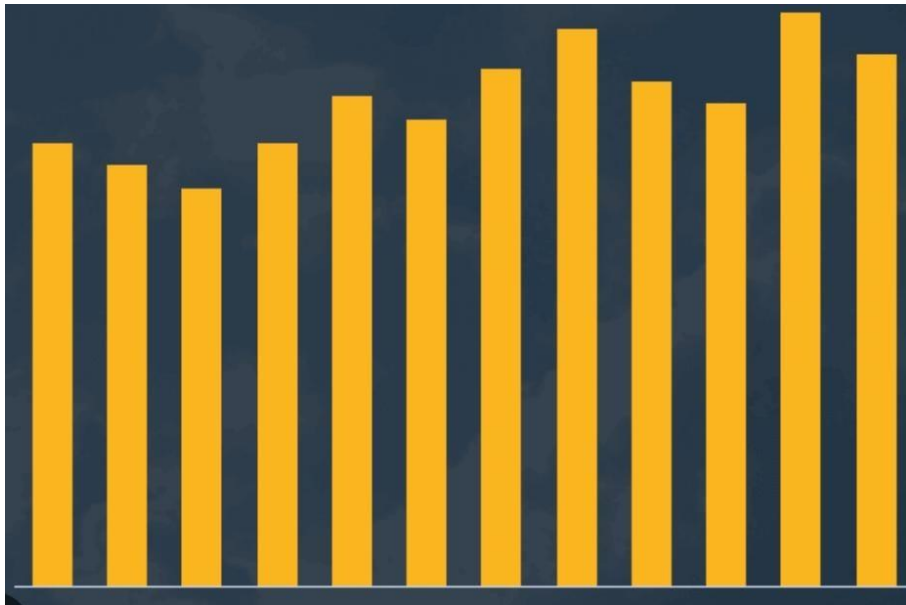
```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
traffic_data=[5.,25.,50.,20.,]
```

```
plt.barh(range(len(Unique visits)),Date)
```

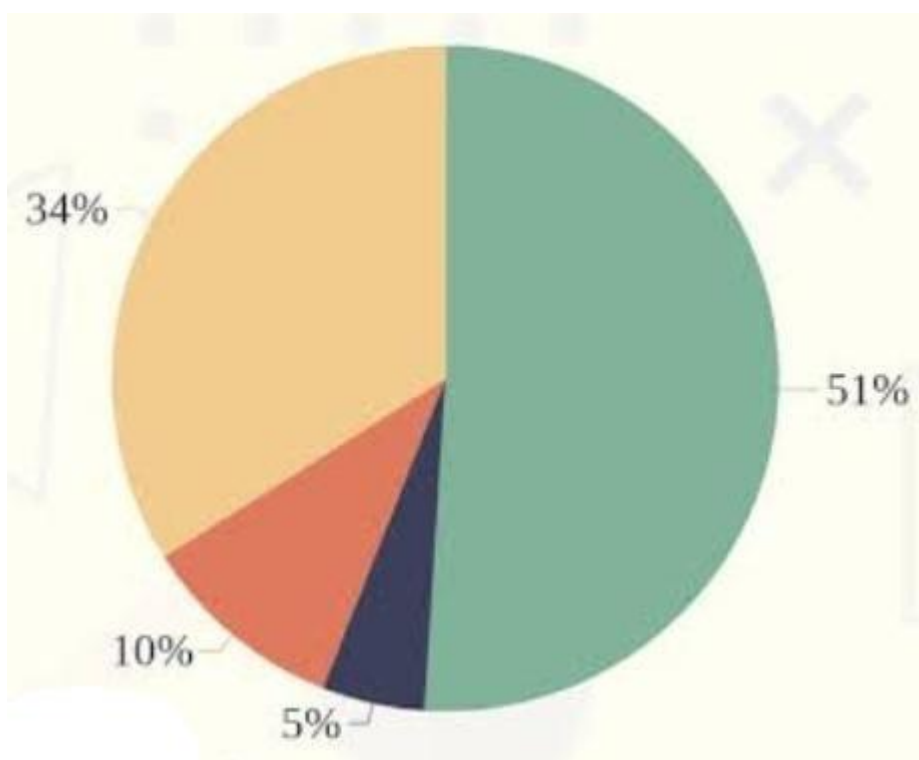
```
plt.show()
```



BARGRAPH AND PIE CHARTS:

```
Country_cases.plot(kind = "pie", ylabel = "", title = "Days in a week with network usage", startangle = 0)
```

```
plt.show()
```



```
Total =
traffic_data.groupby('First.Time.Visits')['Returning.Visits'].sum().sort_values(ascending=False).head(
10)

print(Total)
```

```

Jupyter Naan mudhalvan Last Checkpoint: 6 hours ago
File Edit View Run Kernel Settings Help Trusted
4 5 Thursday 9-18-2014 3,316 2,366

First.Time.Visits Returning.Visits
0 1,430 152
1 2,297 231
2 2,352 278
3 2,327 287
4 2,130 236

[6]: Total = traffic_data.groupby('First.Time.Visits')['Returning.Visits'].sum().sort_values(ascending=False).head(10)
print(Total)

First.Time.Visits
4,469 998
3,998 993
4,237 987
4,452 987
4,500 979
4,010 973
4,369 969
4,120 961
3,988 957
4,465 956
Name: Returning.Visits, dtype: object

In [ ]: 
```

```
import pandas as pd

traffic_data= pd.read_csv(r"C:\Users\tarun\OneDrive\Desktop\daily-website-visitors.csv")

print(traffic_data.head())

print(traffic_data.info())
```

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2167 entries, 0 to 2166
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype  
---  -
0   Row              2167 non-null  int64  
1   Day              2167 non-null  object  
2   Day.Of.Week      2167 non-null  int64  
3   Date             2167 non-null  object  
4   Page.Loads       2167 non-null  object  
5   Unique.Visits    2167 non-null  object  
6   First.Time.Visits 2167 non-null  object  
7   Returning.Visits  2167 non-null  object  
dtypes: int64(2), object(6)
memory usage: 135.6+ KB
None
```

Output Image:

The screenshot displays a JupyterLab environment with a code editor and an output area. The code editor contains the following Python code:

```
[7]: import pandas as pd
traffic_data = pd.read_csv(r"C:\Users\tarun\OneDrive\Desktop\daily-website-visitors.csv")
print(traffic_data.head())
print(traffic_data.info())
```

The output area shows the result of the code execution:

```
3      2,327      287
4      2,130      236
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2167 entries, 0 to 2166
Data columns (total 8 columns):
 #   Column              Non-Null Count  Dtype  
---  --
 0   Row                 2167 non-null  int64  
 1   Day                 2167 non-null  object  
 2   Day.Of.Week         2167 non-null  int64  
 3   Date                2167 non-null  object  
 4   Page.Loads          2167 non-null  object  
 5   Unique.Visits       2167 non-null  object  
 6   First.Time.Visits   2167 non-null  object  
 7   Returning.Visits    2167 non-null  object  
dtypes: int64(2), object(6)
memory usage: 135.6+ KB
None
```

The second code cell contains the following Python code:

```
[6]: Total = traffic_data.groupby('First.Time.Visits')['Returning.Visits'].sum().sort_values(ascending=False).head(10)
print(Total)
```

The output area shows the result of the code execution:

```
First.Time.Visits
4,469      998
3,998      993
4,237      987
```

The JupyterLab interface includes a top bar with the Jupyter logo, the username 'Naan mudhalvan', and the text 'Last Checkpoint: 6 hours ago'. The top bar also contains a menu bar with 'File', 'Edit', 'View', 'Run', 'Kernel', 'Settings', and 'Help'. The top bar also contains a 'Trusted' status indicator. The top bar also contains a 'JupyterLab' logo and a 'Python 3 (ipykernel)' label. The bottom bar shows the system tray with a weather icon (26°C, Mostly cloudy), a search bar, and various system icons (network, volume, battery, etc.). The bottom bar also shows the date and time (01-11-2023, 21:49).